

Summary of the data lifecycle

You've learned that the data lifecycle is the sequence of stages that data experiences, including: plan, capture, manage, analyze, archive, and destroy. Understanding the way data moves through the cycle can help you work effectively with a team, manage projects, and troubleshoot issues quickly. In this reading, you'll review each stage of the data lifecycle, and explore an example of how the cycle works in an organization.

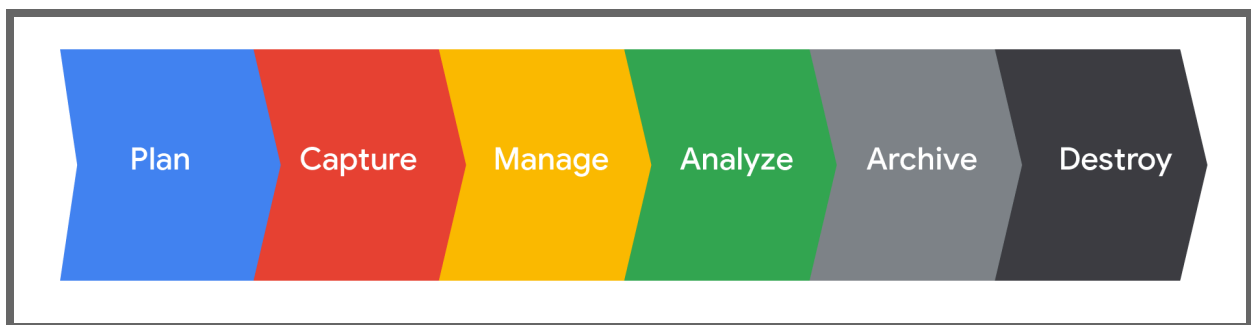
Phases of the data lifecycle

Data lifecycle phases mark the journey that data takes from start to finish of any data project. Here's a list of all the phases and what they mean:

- **Plan:** The planning stage begins before you start any analysis, and involves answering a business question or meeting an objective.

Pro tip: The planning stage should include deciding what type of data to collect, how it will be managed, and who is responsible for each stage of the lifecycle.

- **Capture:** In the capture phase, data is collected from different sources, and gaps are identified.
- **Manage:** The manage phase ensures that data is properly maintained, and stored safely in a secure location.
- **Analyze:** In the analyze phase, analysts use data to answer the business question or help meet the business objectives.
- **Archive:** In the archive phase, data is stored for later use, and for compliance purposes.
- **Destroy:** In the destroy phase, the data that is no longer useful to the organization is permanently deleted.



Key takeaways

Working with data through each phase of the data lifecycle ensures there's transparency in the process. It also ensures that every member of the team understands their role in the analysis project. Approaching analysis as a cyclical process gives space for iteration and **refinement**.

Resources for more information

These resources provide more information about how data moves through a lifecycle:

- The University of Wisconsin-Madison provides two examples of how data moves through the lifecycle: <https://data.wisc.edu/data-literacy/lifecycle/>
- Shopify provides extended definitions of each phase of the data lifecycle: <https://www.shopify.com/enterprise/data-life-cycle>